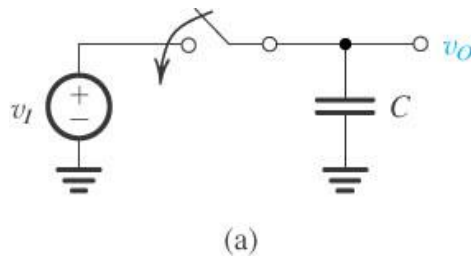


Chapter 5

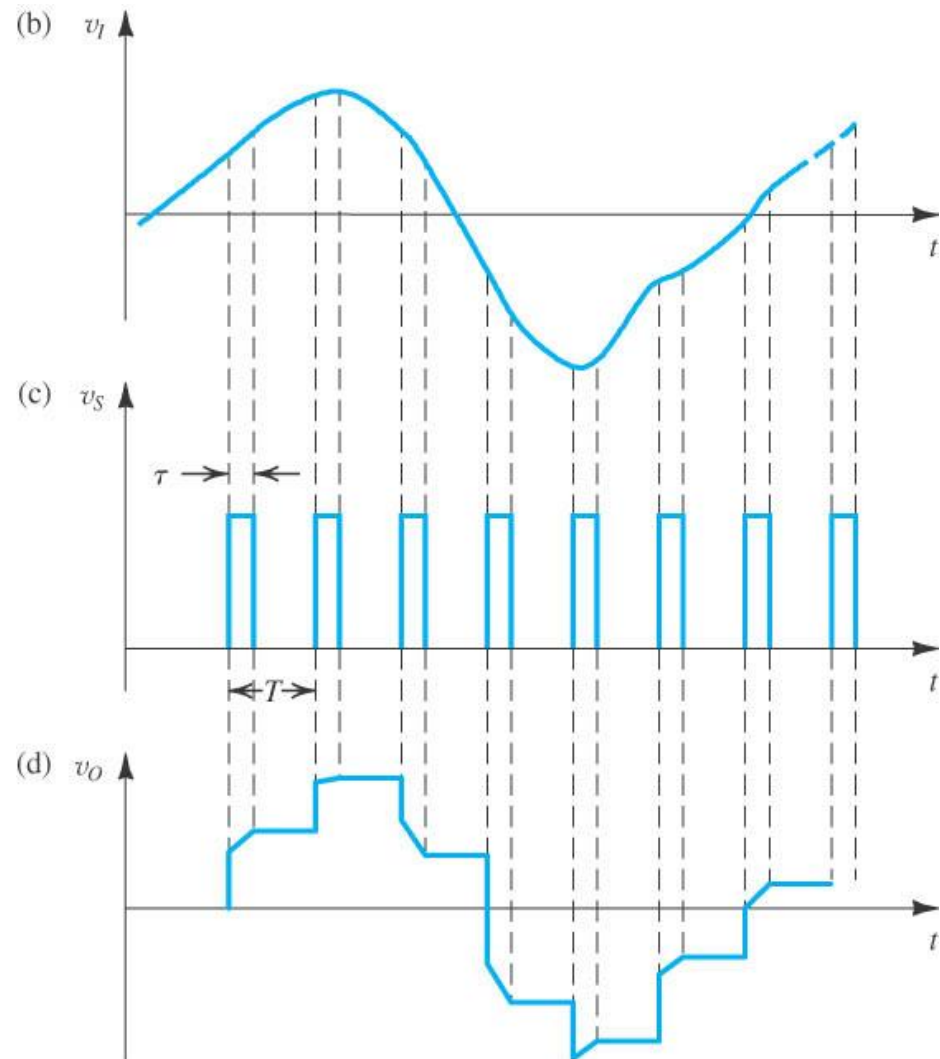
Data-Converter Circuits

A/D and D/A

Most signals are
analog



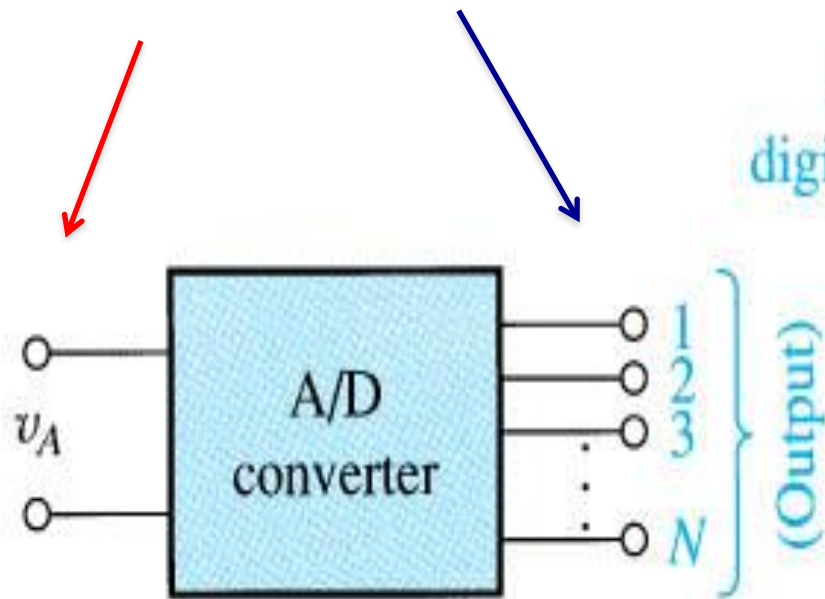
Need to Sample an analog
signal
Then convert to digital by
A/D converter



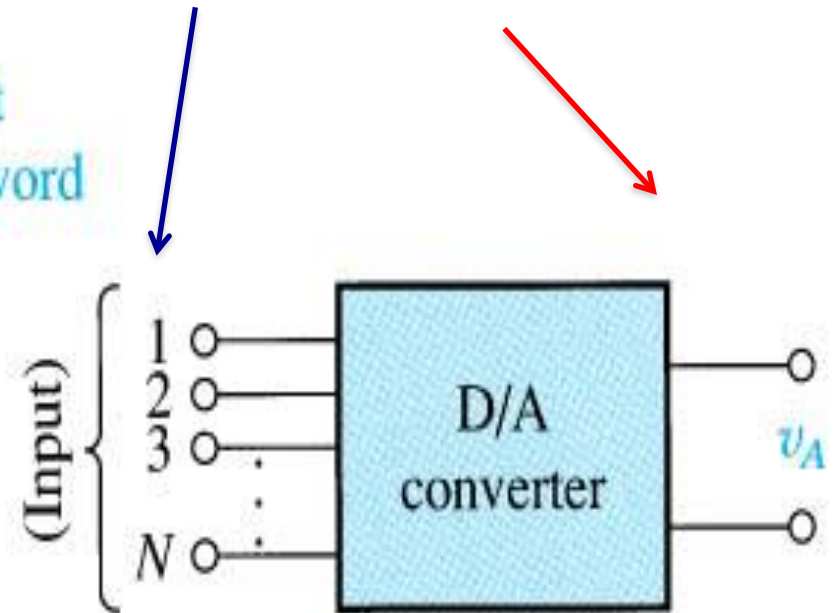
The process of periodically sampling an analog signal. **(a)** Sample-and-hold (S/H) circuit. The switch closes for a small part (τ seconds) of every clock period (T). **(b)** Input signal waveform. **(c)** Sampling signal (control signal for the switch), **(d)** Output signal (to be fed to A/D converter).

A/D converter and D/A Converters

Analog to Digital



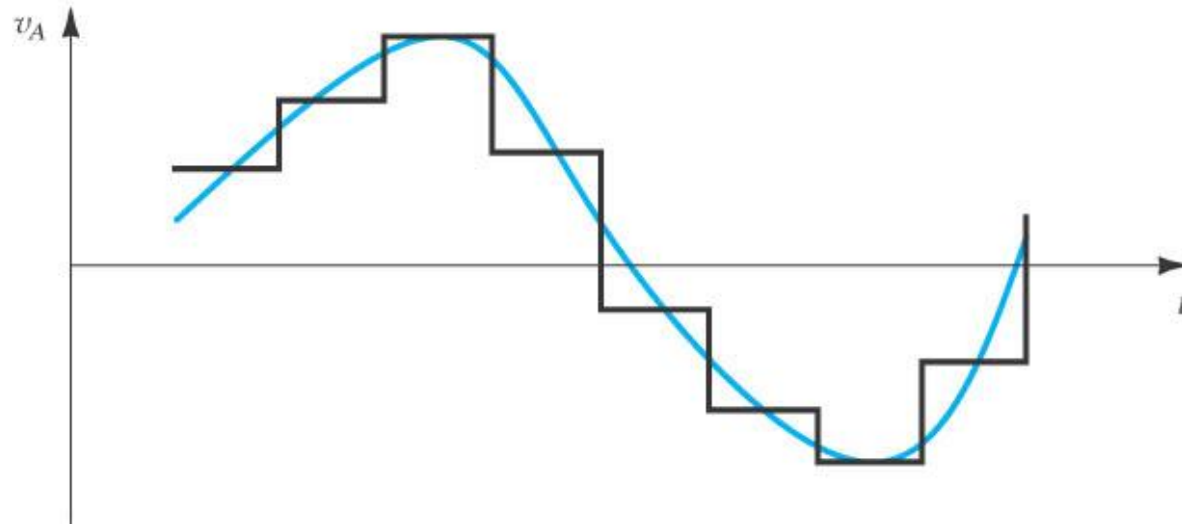
Digital to Analog



N -bit
digital word

D/A Conversion

Normal Output from **digital** domain is staircase
Filtered to produce smooth **Analog** output



Accuracy

There are several ways of specifying accuracy. The most common is called full-scale error, which is normally expressed as a percentage of the converter's full-scale output (%F.S.)

Full-scale error is the maximum deviation of the DAC's output from its expected (ideal) value, expressed as a percentage of full scale. For example, assume that the DAC has an accuracy of $\pm 0.01\%$ F.S. Since this converter has a full-scale output of 9.375 V, this percentage converts To $\pm 0.01\% \times 9.375 \text{ V} = \pm 0.9375 \text{ mV}$

This means that the output of this DAC can, at any time, be off by as much as 0.9375mV from its expected value.

Resolution

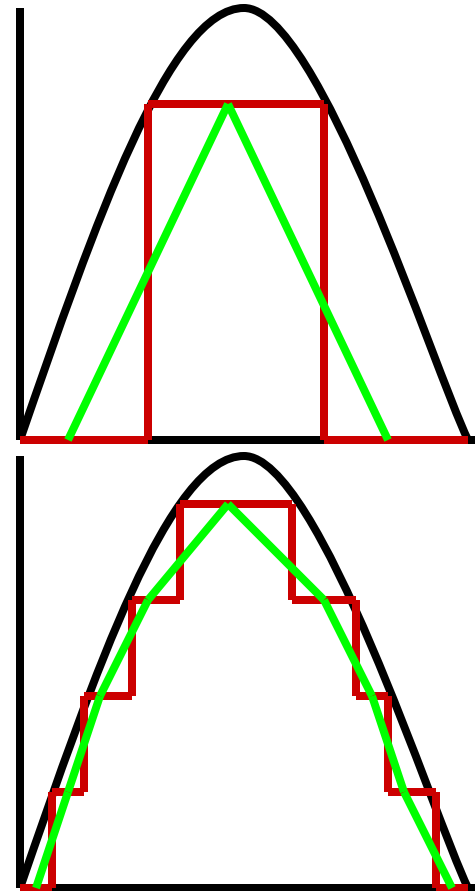
- Smallest analog increment corresponding to 1 LSB change
- An N-bit resolution can resolve 2^N distinct analog levels
- Common DAC has a 8-16 bit resolution

$$\text{Resolution} = V_{LSB} = \frac{V_{ref}}{2^N}$$

where $N = \text{number of bits}$

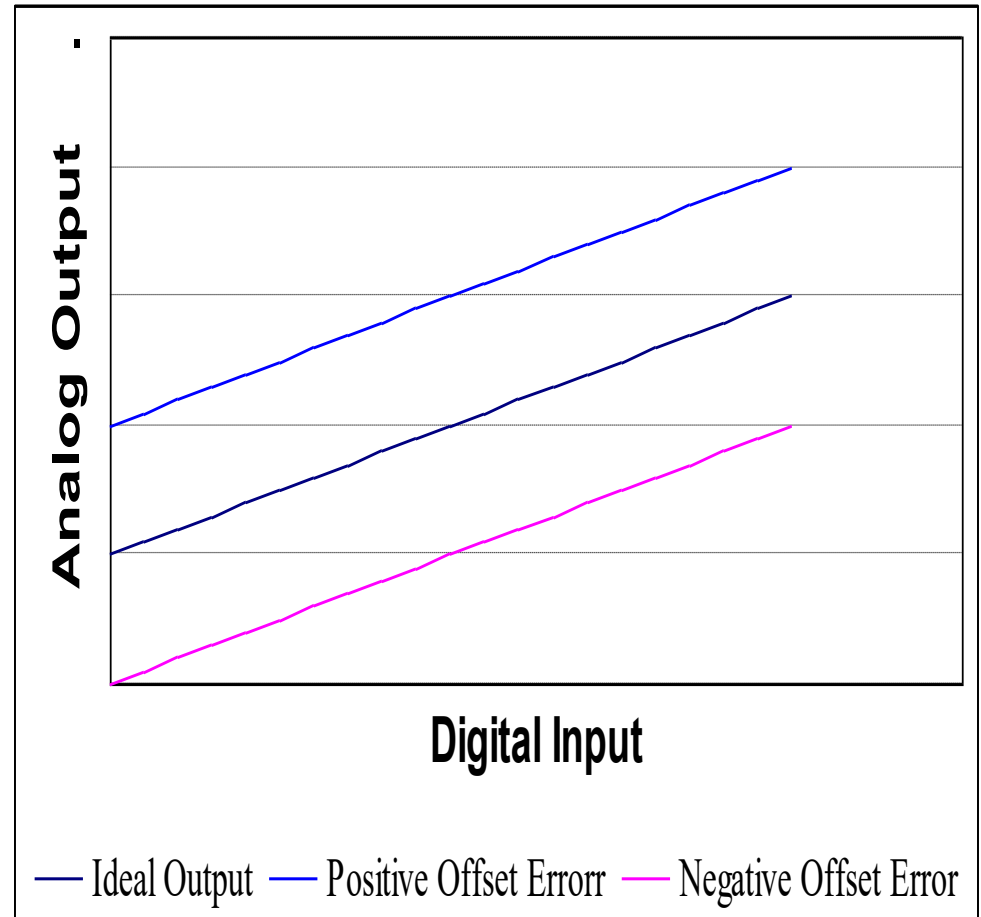
Resolution Error

- ❑ Poor representation of ideal output due to poor resolution
- ❑ Size of voltage divisions affect the resolution



Offset Error

- Occurs when there is a constant offset between the actual output and the ideal output



Offset error - difference between an ideal and actual DAC output when zero digital code applied to the input.

Gain error

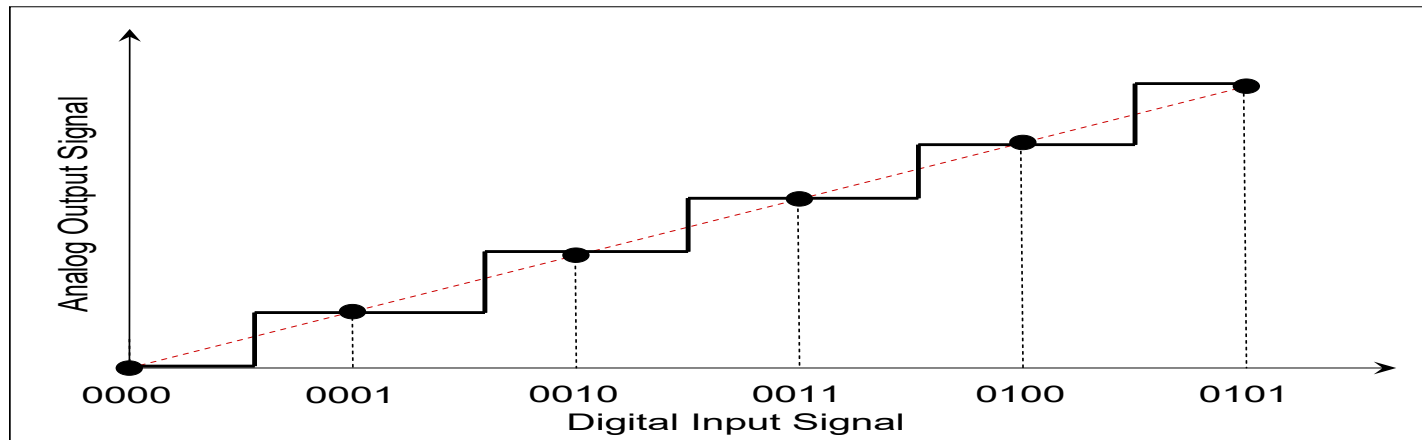
the difference between an ideal and actual output when full scale digital code applied to the input.

A gain will produce output step sizes that are either larger or smaller than the expected LSB.

Gain error causes the output to deviate further from the expected value as the binary input is increased.

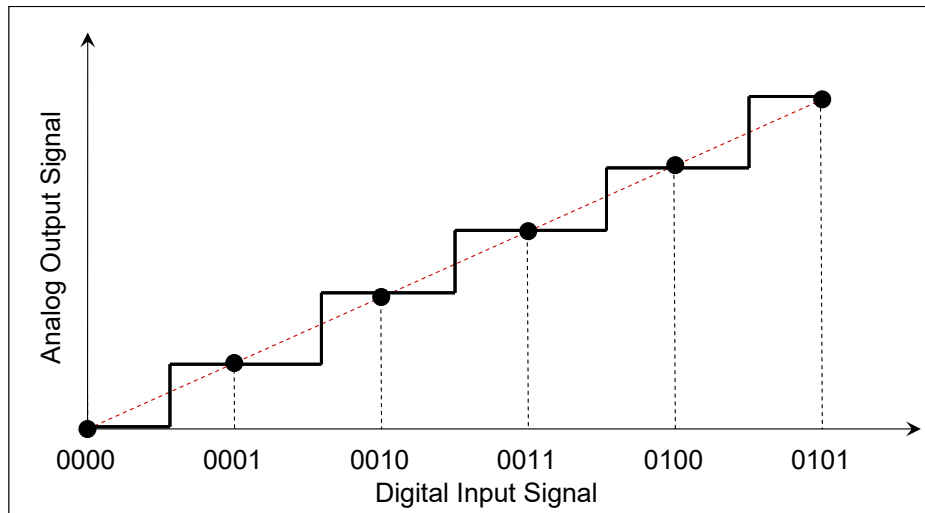
Linearity error

- ❑ The difference between the desired analog output and the actual output over the full range of expected values

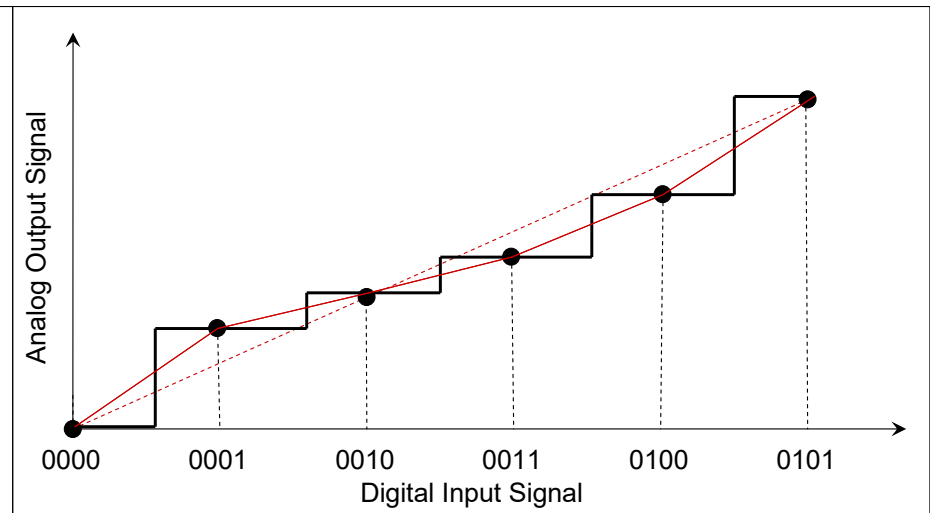


□ Ideally, a DAC should produce a linear relationship between the digital input and analog output

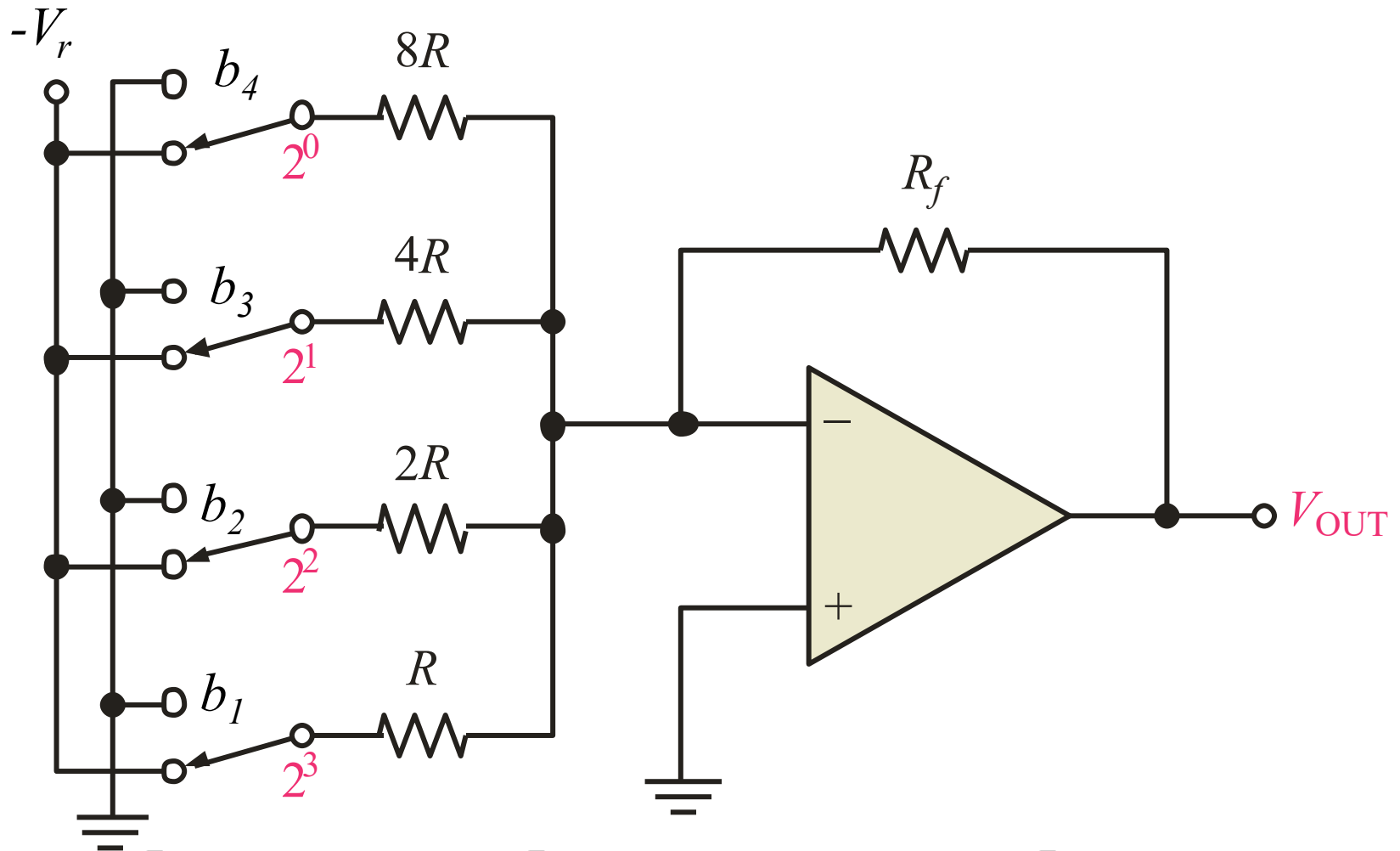
Linearity (Ideal)



Non-Linearity



D/A Converter with binary weighted resistor



$$V_o = \frac{R_f}{2^3 R} b_4 V_r + \frac{R_f}{2^2 R} b_3 V_r + \dots + \frac{R_f}{R} b_1 V_r$$

$$V_o = \frac{2R_f V_r}{R} (b_1 2^{-1} + b_2 2^{-2} + \dots + b_n 2^{-n})$$

- **Advantages**

- Simple Construction Analysis
- Fast Conversion

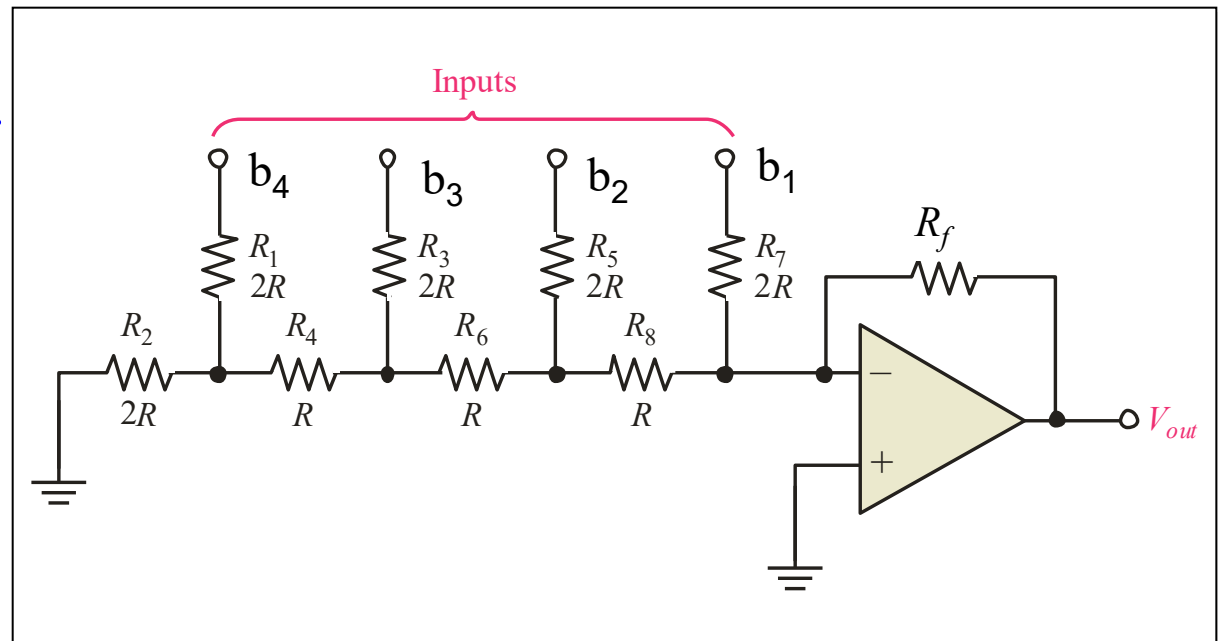
- **Disadvantages**

- Requires large range of resistors (2048:1 for 12-bit DAC) with necessary high precision for low resistors
- Requires low switch resistances in transistors
- Can be expensive. Therefore, usually limited to 8-bit resolution.

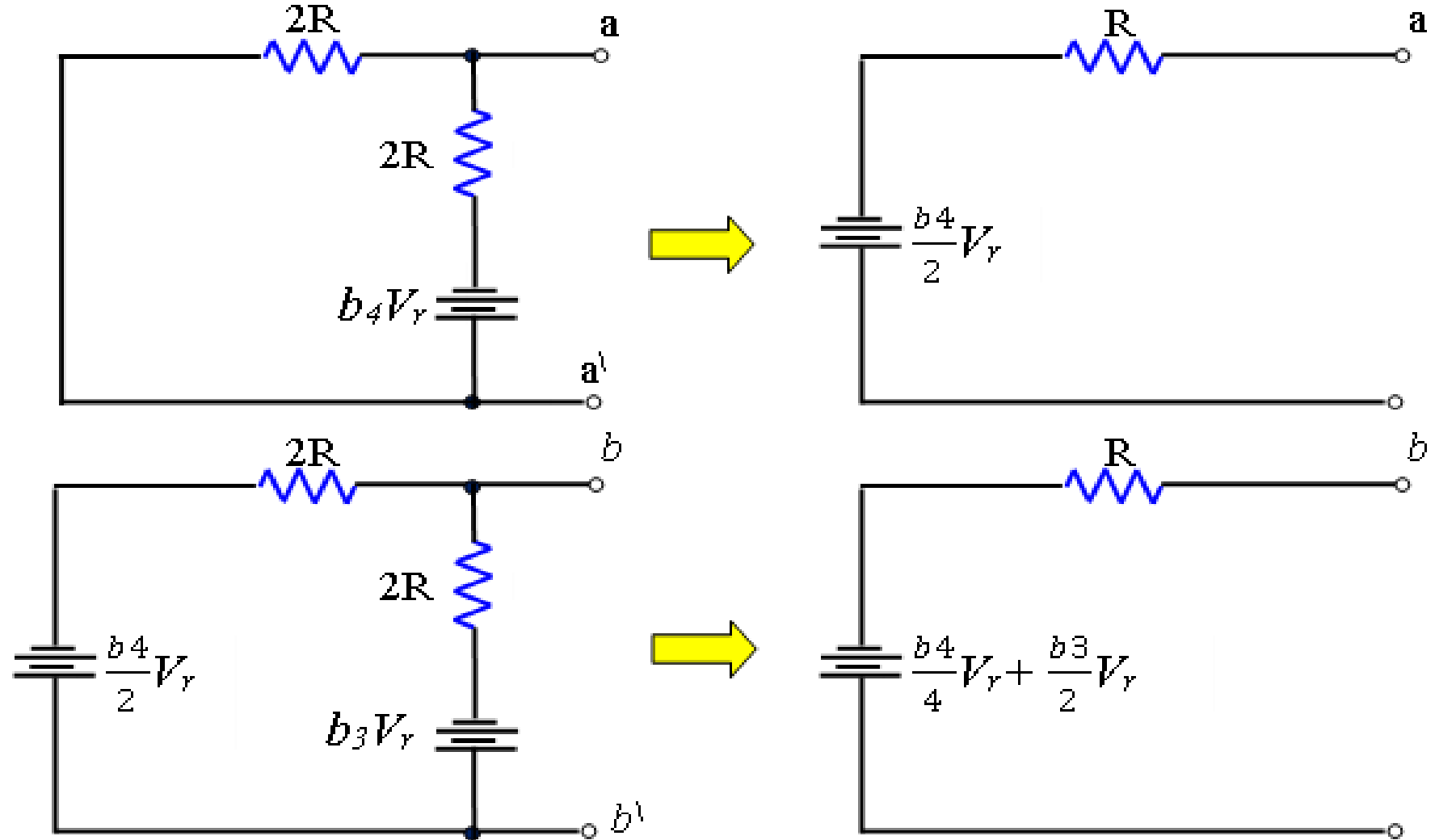
$R/2R$ Ladder DAC

A more widely used method for D/A conversion is the $R/2R$ ladder. The gain for b_1 is -1. Each successive input has a gain that is half of previous one. The output represents a weighted sum of all of the inputs (similar to the scaling adder).

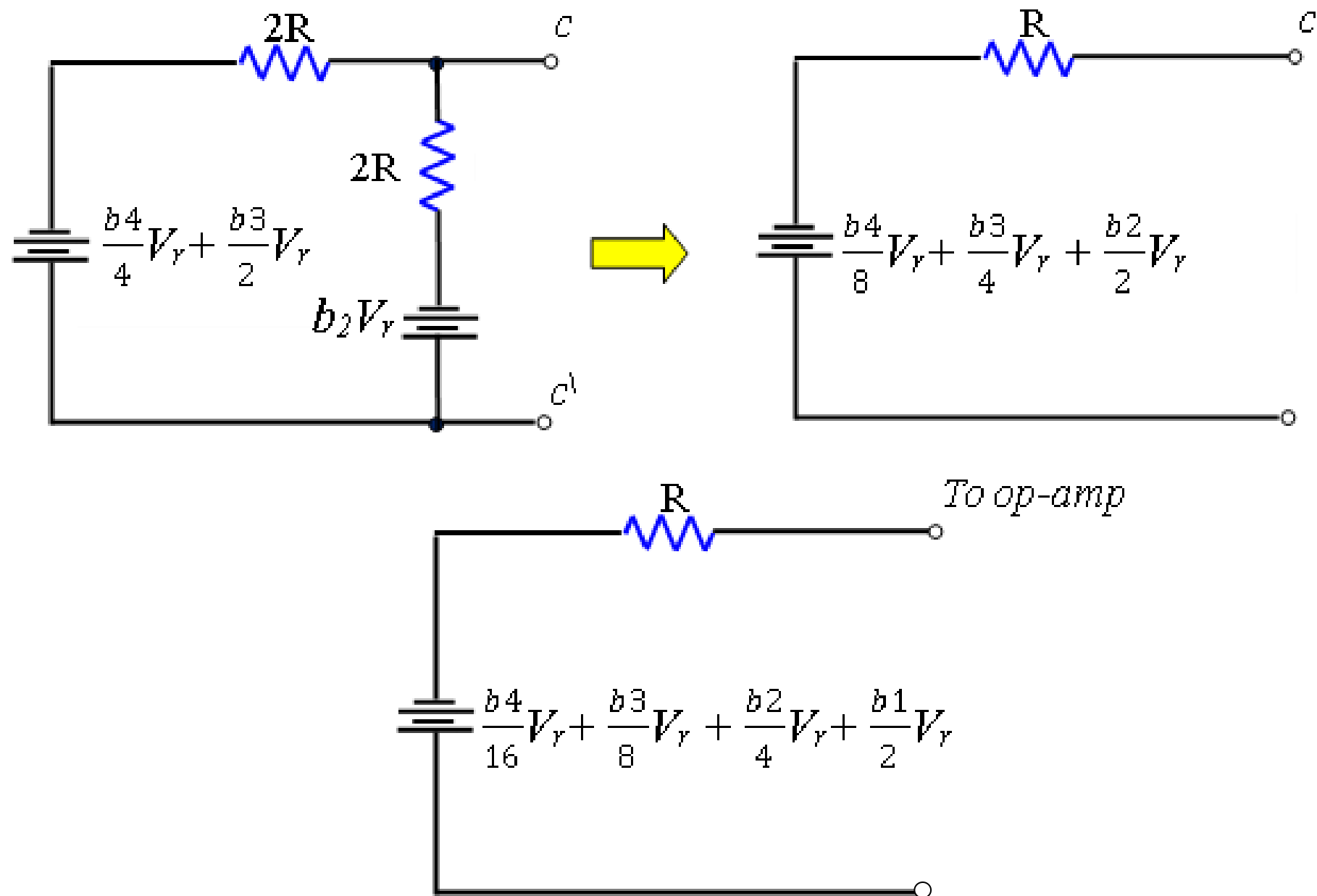
An advantage of the $R/2R$ ladder is that only two values of resistors are required to implement the circuit.



$$V_{out} = \frac{R_f V_r}{R} (b_1 2^{-1} + b_2 2^{-2} + b_3 2^{-3} + b_4 2^{-4})$$



$$\begin{aligned}
 & \left(b_3 V_r - \frac{b_4}{2} V_r \right) (2R/4R) - b_3 V_r \\
 &= \left(\frac{b_3}{2} V_r - \frac{b_4}{4} V_r \right) - b_3 V_r = - \left(\frac{b_4}{4} V_r + \frac{b_3}{2} V_r \right)
 \end{aligned}$$



$$V_{out} = \frac{R_f V_r}{R} (b_1 2^{-1} + b_2 2^{-2} + b_3 2^{-3} + b_4 2^{-4})$$

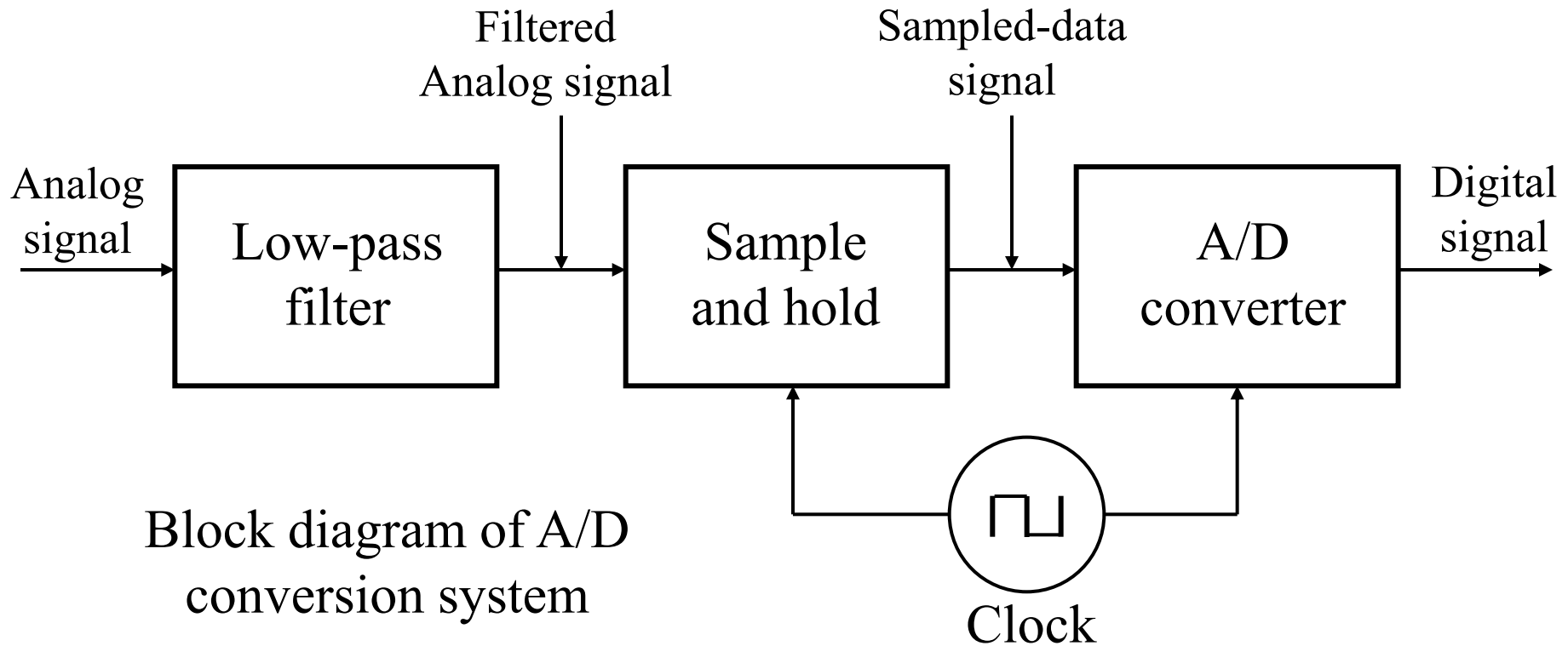
- Advantages

- Only two resistor values (R and 2R)
- Does not require high precision resistors

- Disadvantage

- Lower conversion speed than binary weighted DAC

A/D Converters



Shannon's sampling theorem states that the signal can be reproduced by samples taken at intervals no greater than $1/2f_{in}$ apart.

$$f_{sa} \geq 2f_{in}$$

Shannon's sampling theorem

Shannon's sampling theorem states that the signal can be reproduced by samples taken at intervals no greater than $1/2f_{in}$ *apart*.

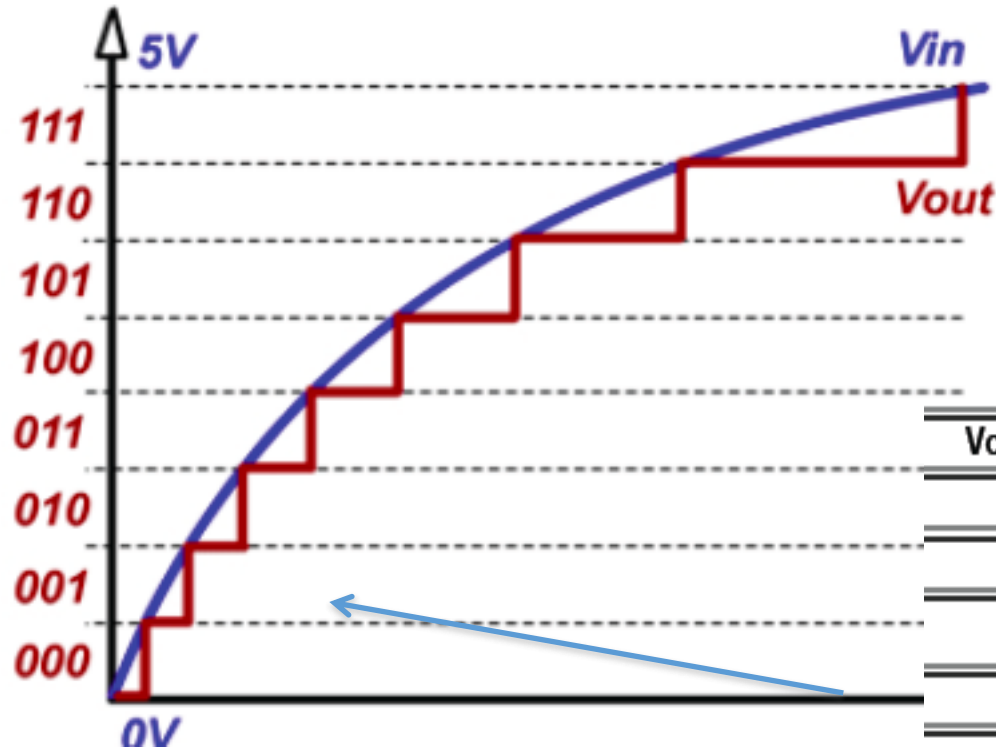
$$f_{sa} \geq 2f_{in}$$

Aliasing

Occurs when the input signal is changing much faster than the sample rate.

For example, a 2 kHz sine wave being sampled at 1.5 kHz would be reconstructed as a 500 Hz (the aliased signal) sine wave.

3-bit conversion example



Quantization levels

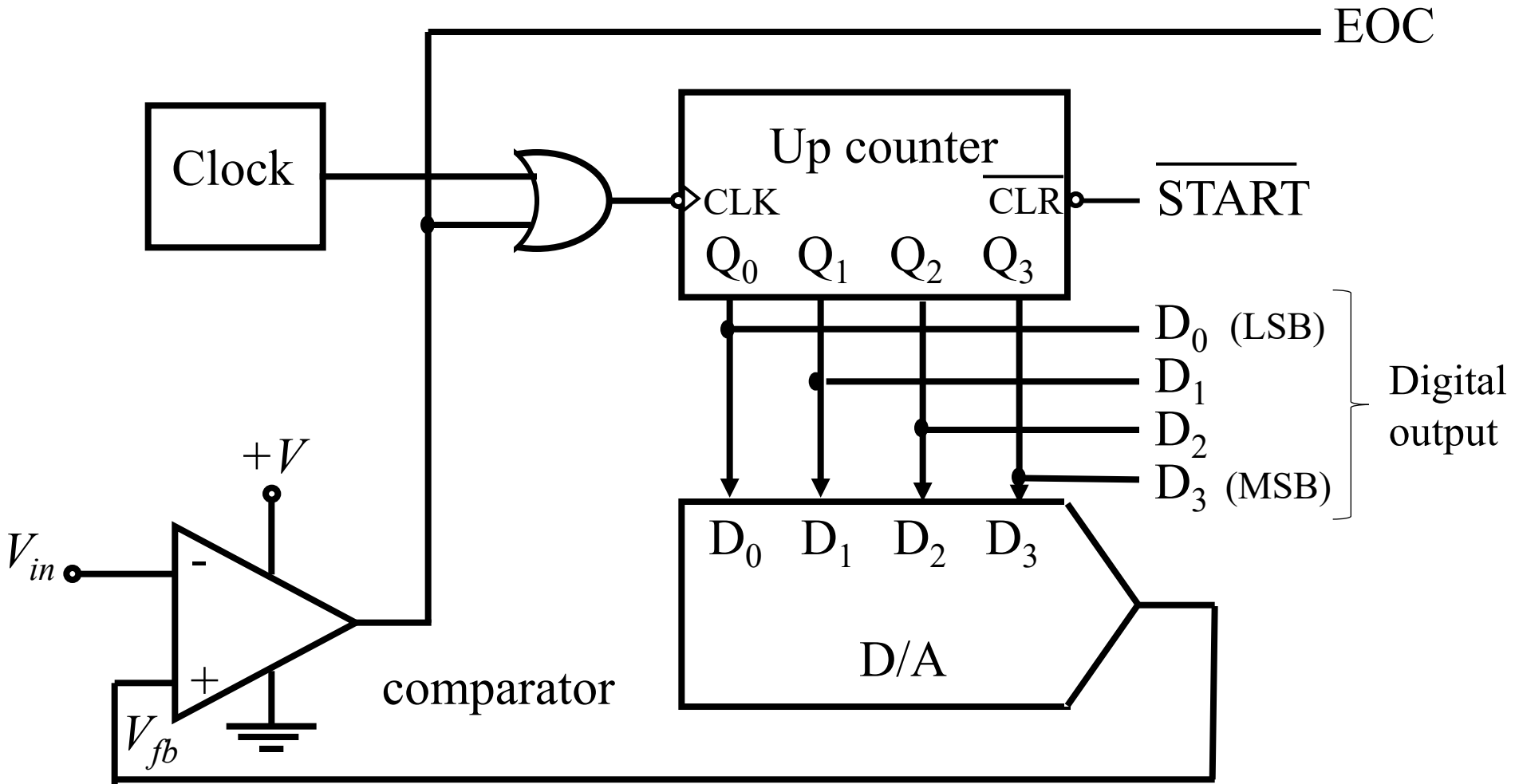
Voltage levels [V]	Binary representation
0-0.62	000
0.621-1.25	001
1.251-1.87	010
1.871-2.5	011
2.51-3.12	100
3.121-3.75	101
3.751-4.37	110
4.371-5.00	111

Eg 5V divided into 8 levels – each 0.625

$$\text{Step size} = V_{FS} / 2^n$$

The difference between the analog signal amplitude and the nearest standard digital level is called **quantization error**

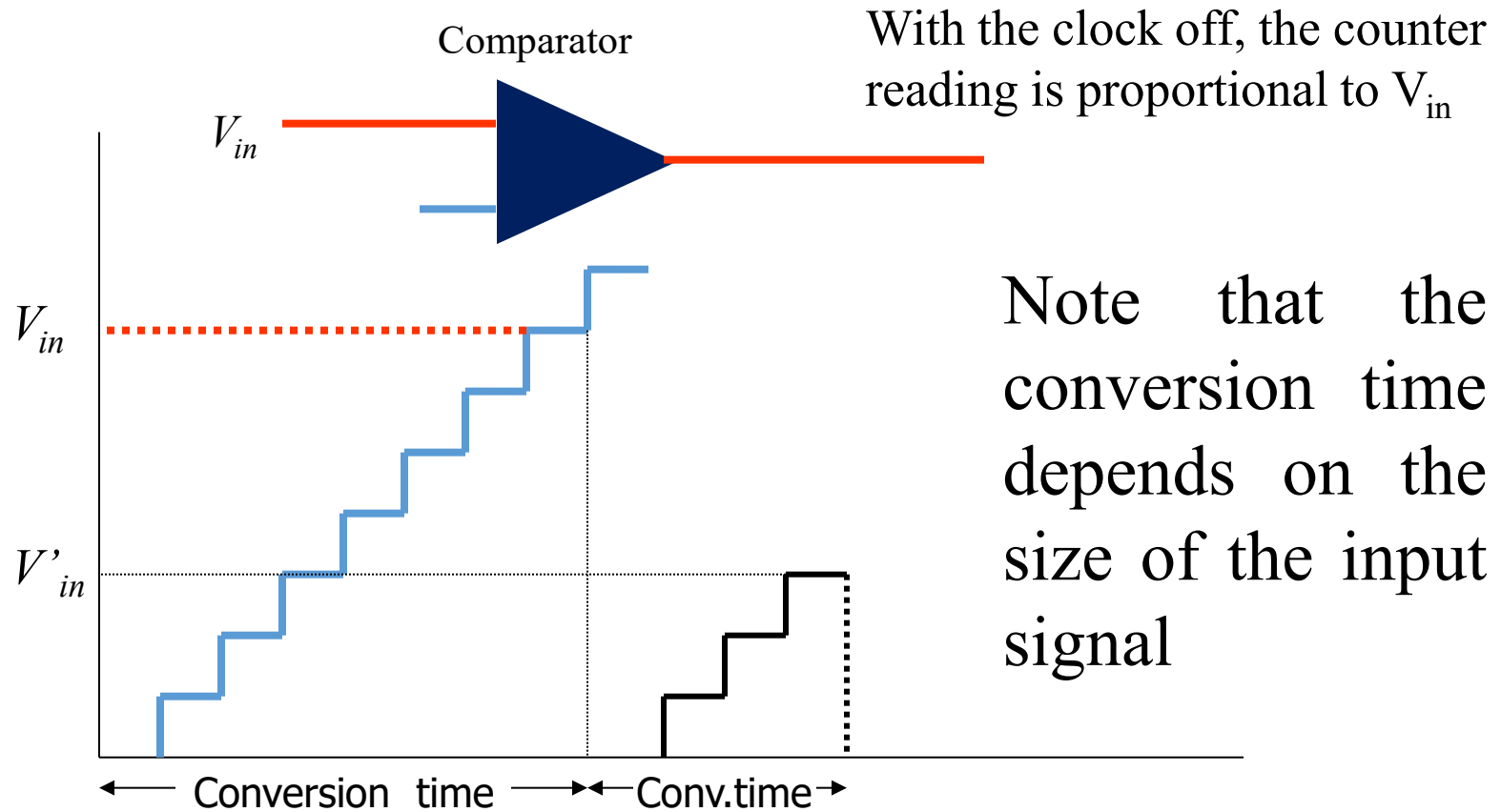
Counter (or ramp) A/D converter



- The $\overline{\text{START}}$ input is driven low. This clears the counters, causing the output of the D/A converter $V_{fb} = 0$
- Because $V_{in} > V_{fb}$, the output of the comparator is driven low, enabling the OR gate to pass clock pulses to the up counter. The output of the comparator is also used to indicate that a conversion is now in progress ($\text{EOC} = 0$)
- The counter increments once for each falling edge of the clock signal, forcing the output of the D/A converter to increase by 1 LSB on each falling clock edge. This action continues until the output of the D/A converter is greater than the input voltage being sampled.

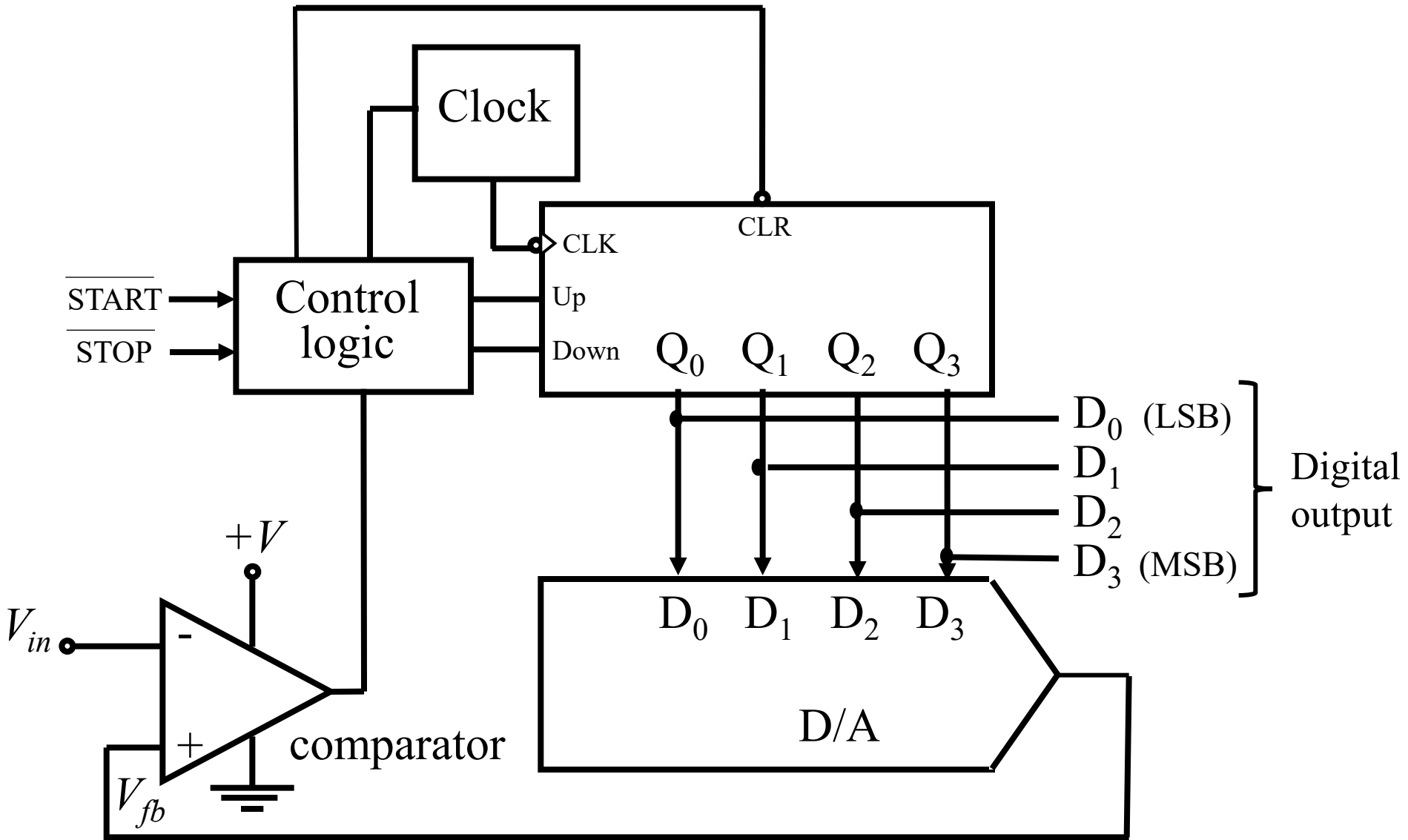
➤ Once $V_{fb} > V_{in}$, the output of the comparator is driven high, which disables the clock input of the counter. A high logic state on the EOC output indicates that the conversion is complete and the output count is proportional to the analog input voltage. Another conversion may be initiated by pulsing the START input low.

As the counter counts, its output to the DAC generates a staircase ramp to the comparator.



With a counter type A/D, if the signal is varying rapidly, the counter must count up and reset before each cycle can begin, making it difficult to follow the signal.

Tracking A/D converter

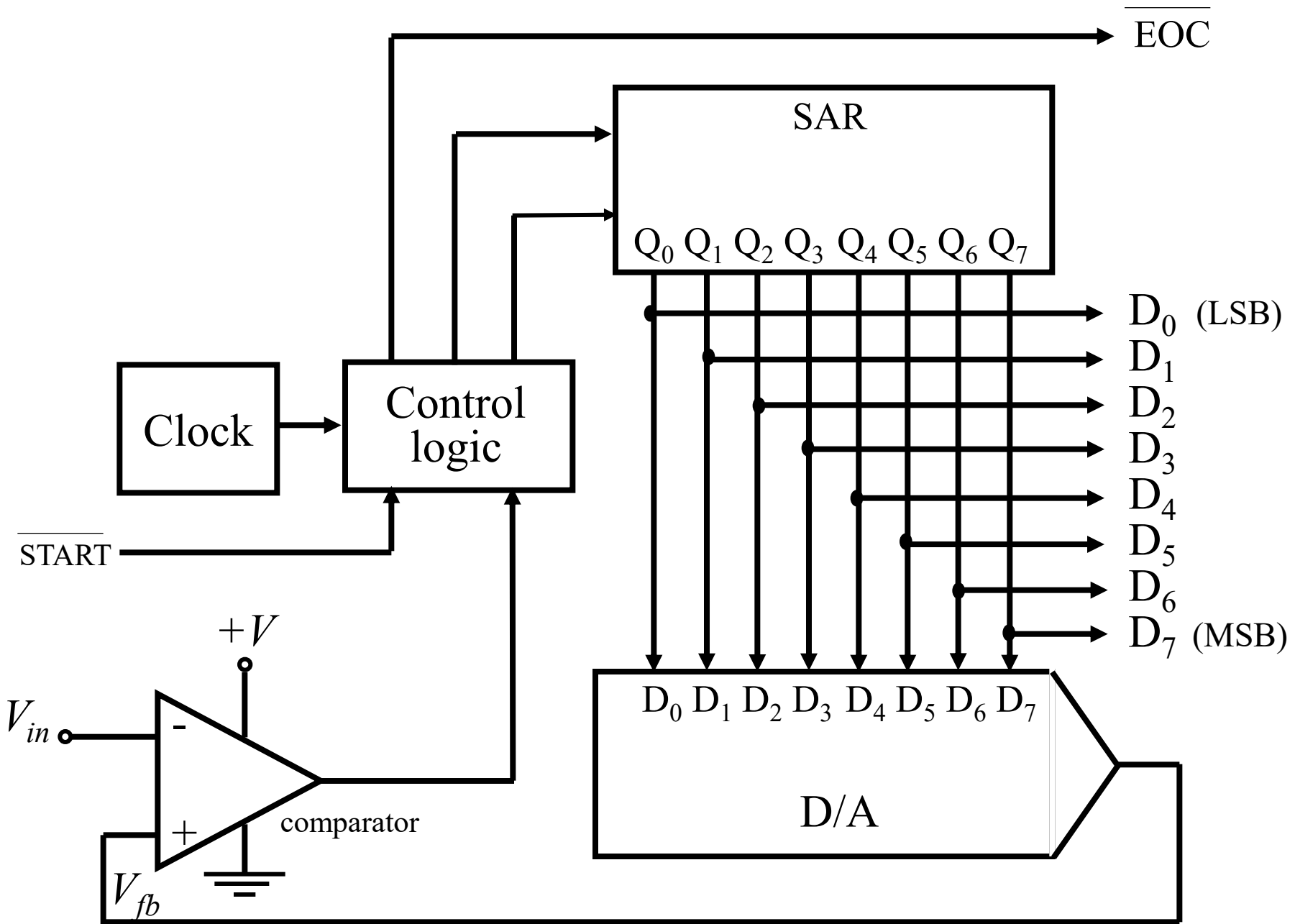


- Once a conversion is started, the up/down counter begins counting up. When the output of the D/A just exceeds the input voltage, the output of the comparator goes high, causing the counter to begin counting down.
- If the input is time varying, the output of the counter will track the input voltage as it increases and decreases; thus the term tracking converter.
- Tracking converters are sometimes called continuous converters because they do not stop after a given conversion is complete.

- Tracking converters have much faster conversion times than ramp converters, making them better suited for higher-speed applications.

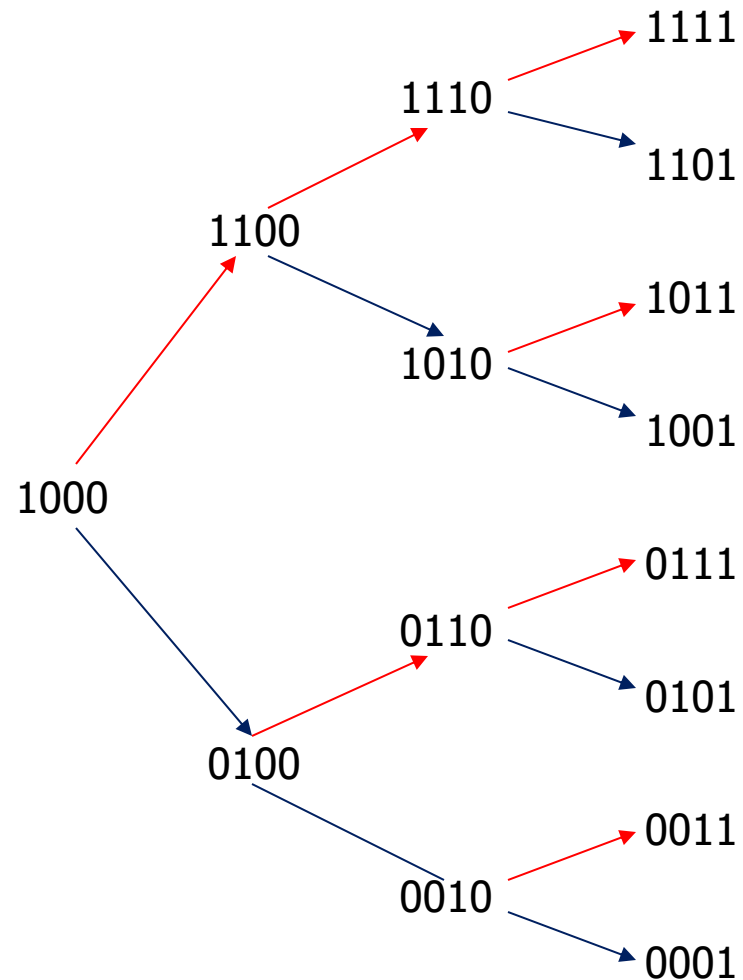
Successive approximation A/D converter

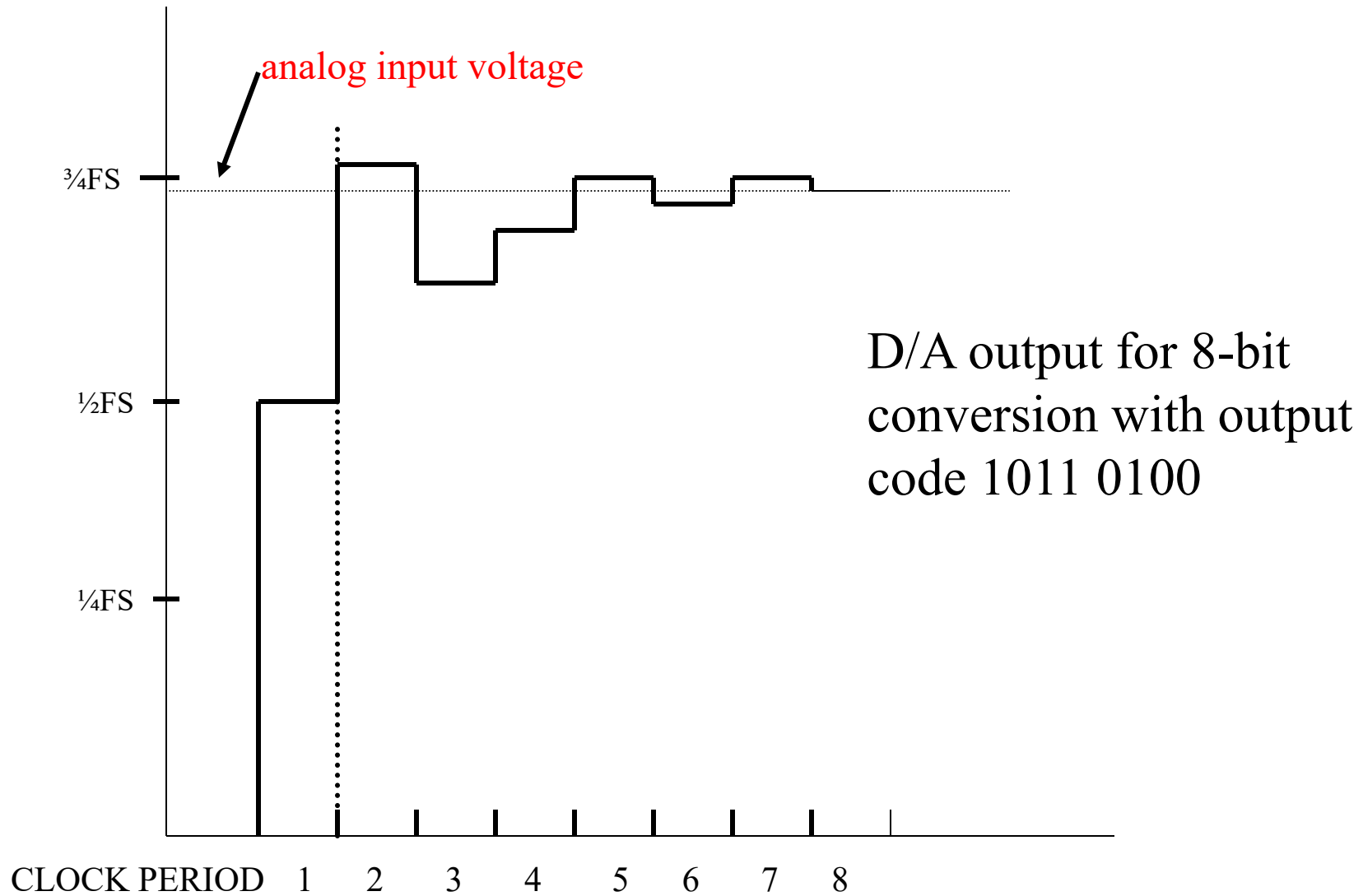
- Much of the operation of a successive approximation A/D converter depends on the control logic.
- The SAR (Successive Approximation Register) is used to produce estimations of the input voltage, beginning with MSB and working down to the LSB, one bit position at a time.
- The output of the internal D/A converter is compared with the input voltage after each internal state change.
- If the setting of a particular bit results in $V_{fb} > V_{in}$, then that bit is cleared. If the setting of a particular bit results in $V_{fb} < V_{in}$, that bit is latched high.



Successive approximation search tree for a 4-bit A/D

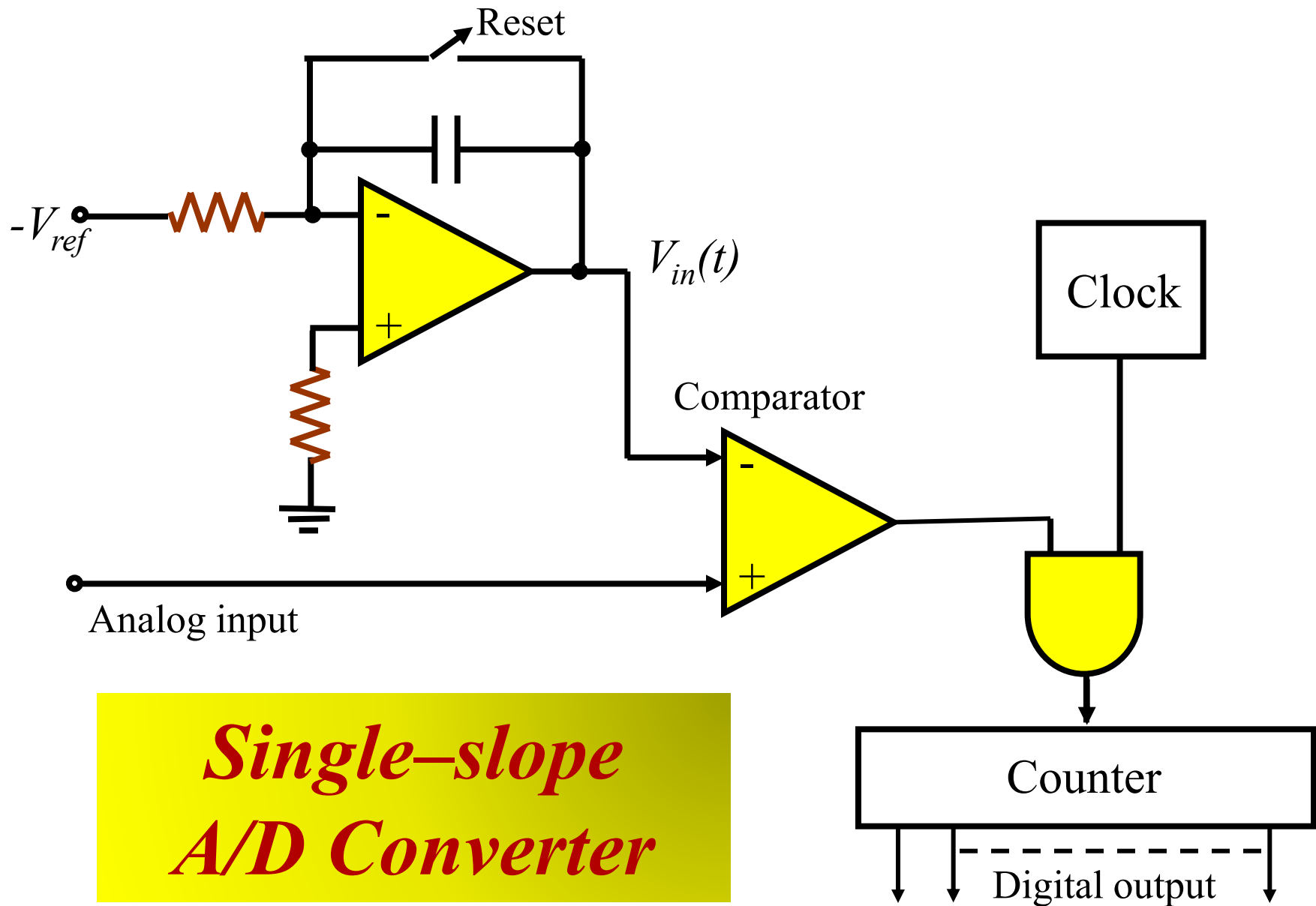
D/A output compared
with V_{in} to see if
larger or smaller





Successive approximation converters are reasonably priced for large bit values, i.e. 10, 12 and even 16 bit converters. Their conversion times, typically $\sim 10\text{-}20\ \mu\text{s}$, are adequate for most laboratory functions.

Integrating A/D Converters



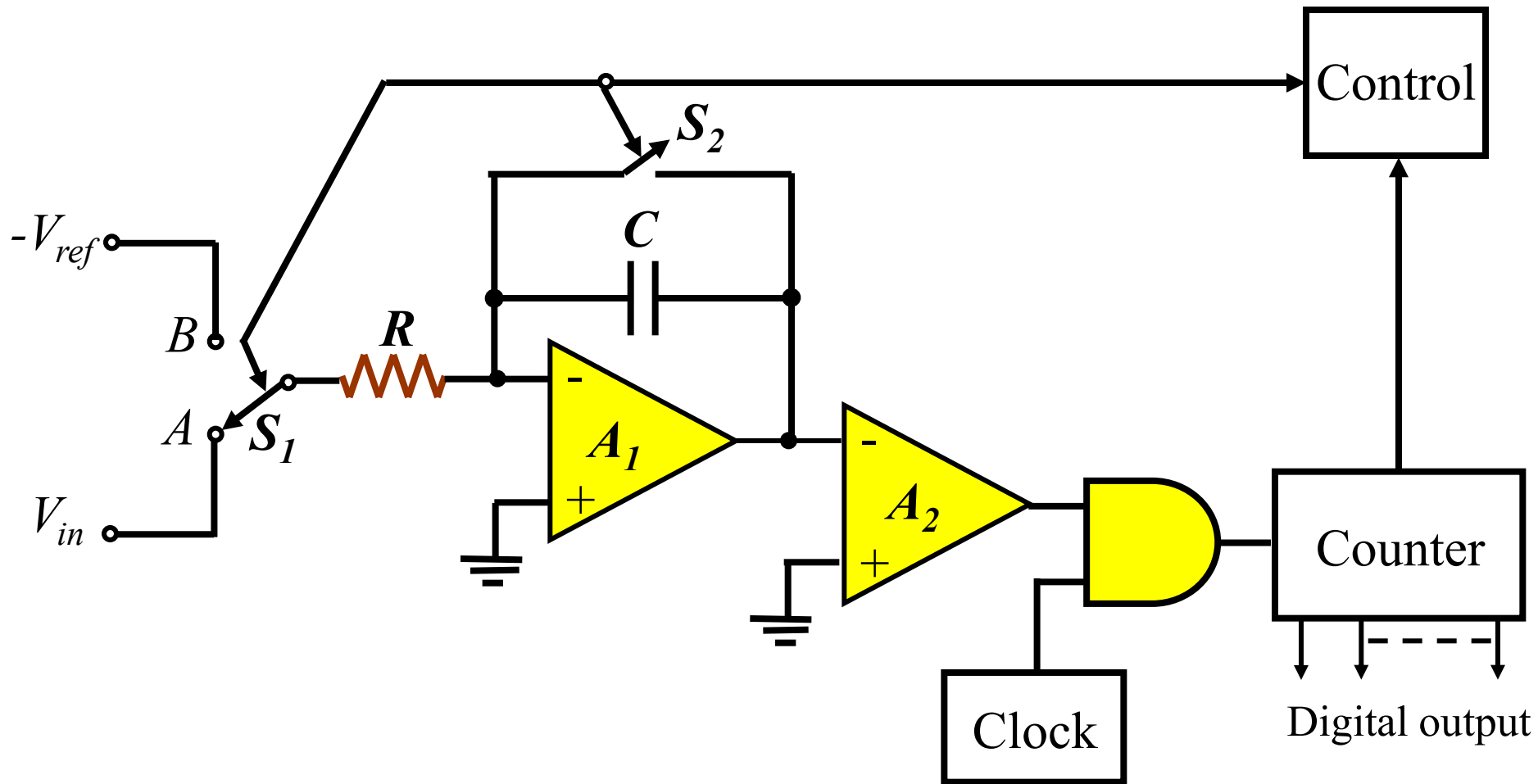
- *Integration is initiated with an input voltage $-V_{ref}$, and the digital counter is started at the same time.*
- *The output of the integrator at time t can be expressed as:*

$$v_{in}(t) = -\frac{1}{RC} \int_0^t (-V_{ref}) dt = \frac{V_{ref}}{RC} t$$

- *When the integrator output voltage reaches the level of the analog signal, the comparator changes states. By means of the associated logic, the counter is inhibited.*
- *The time it takes the integrator output to reach the level of the analog signal is proportional to the level of the analog signal.*

- *The value of the digital word contained in the counter is proportional to the level of the analog signal.*
- *The single-slop converter suffers from possible component-value errors.*
- *The level of the integrator output voltage is a function of the RC product*
- *Changes of capacitance and/or resistance with temperature will affect the integrator output and introduce error.*

Dual-slope A/D Converter



- When conversion is initialized, the switch S_1 is connected to V_{in} (position A) which is applied to the op amp integrator. The integrator output is applied to the comparator
- As conversion is initiated, the control logic enables the clock which then sends pulses to the counter, which counts the pulses until it overflows at time t_2 .

$$V_1(t) = -\frac{1}{RC} \int_{t_1}^t V_{in} dt \quad t_1 \leq t \leq t_2$$

$$V_1(t) = -\frac{V_{in}}{RC} (t - t_1)$$

$$\text{At } t = t_2$$

$$V_1(t) = -\frac{V_{in}}{RC} (t_2 - t_1)$$

$$\text{If } T_1 = t_2 - t_1$$

$$V_1(t) = -\frac{V_{in} T_1}{RC}$$

- As the counter overflows, a “carry” pulse from the counter causes the control logic circuit to set switch S_1 to position B.
- The negative reference voltage removes the charge stored in the integrator until the charge becomes zero.
- At this point, the comparator switches states producing a signal that disables the clock and freezes the counter reading.

➤ During the period of discharge, from t_2 to t_3 the voltage $V_2(t)$ at the output of the integrator is given by the expression.

$$V_2(t) = V_1(t_2) + \frac{1}{RC} \int_{t_2}^t V_{ref} dt$$

$$V_2(t) = V_1(t_2) + \frac{1}{RC} V_{ref} (t - t_2)$$

$$\text{At } t = t_3 \rightarrow V_2(t_3) = 0$$

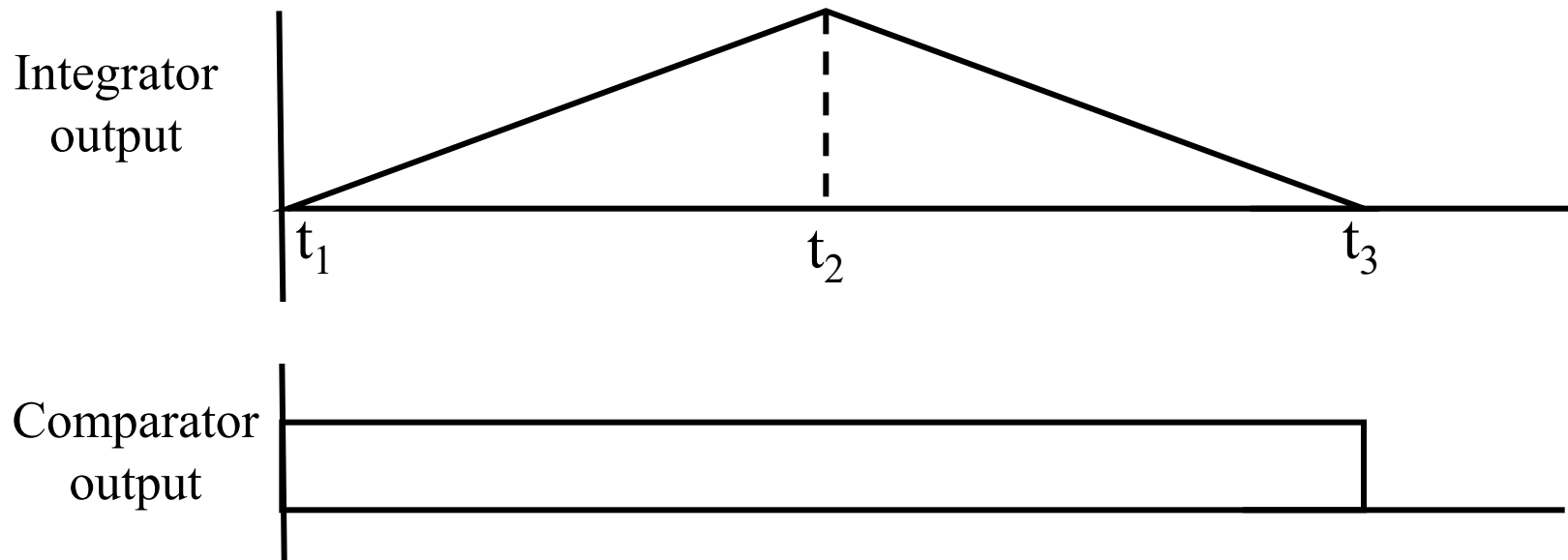
$$\frac{1}{RC} V_{ref} (t_3 - t_2) = \frac{1}{RC} V_{in} T_1$$

$$\text{If } T_2 = t_3 - t_2$$

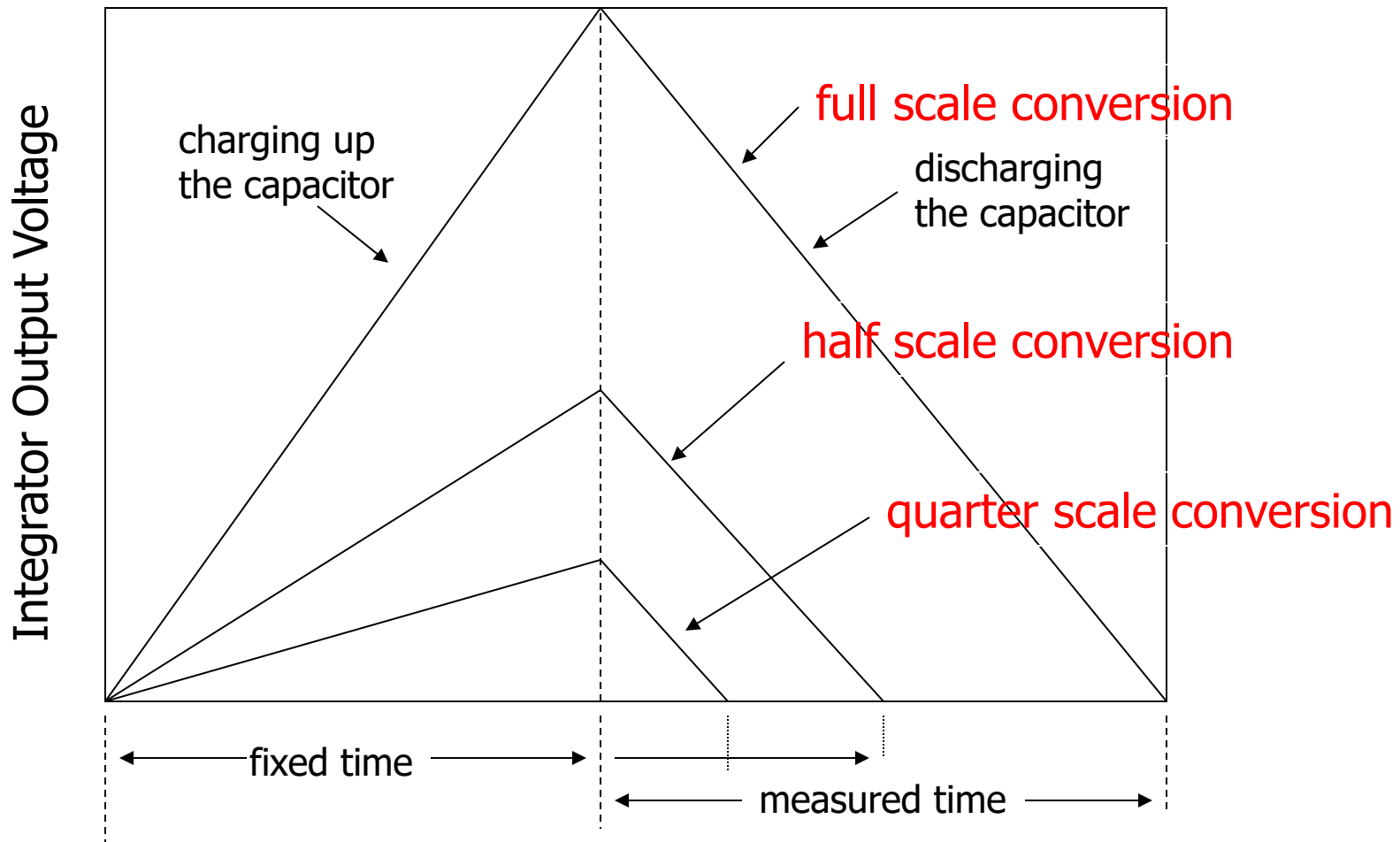
$$V_{ref} T_2 = V_{in} T_1$$

Which equal,

$$\frac{T_2}{T_1} = \frac{V_{in}}{V_{ref}}$$

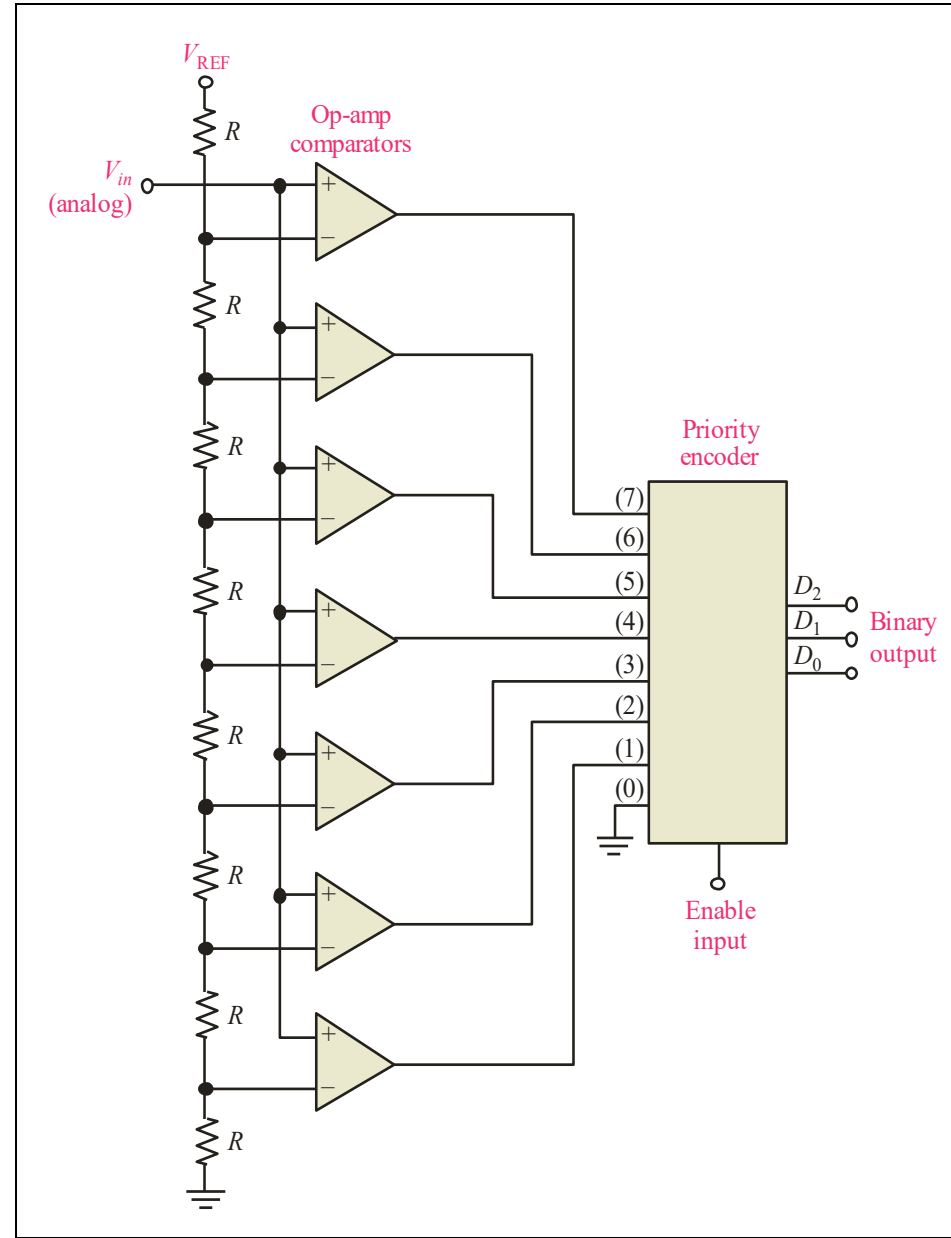


- The operation of this A/D requires 2 voltage slopes, hence the common name DUAL-SLOPE.



Flash Analog to Digital Converter

- Simultaneous or flash analog-to-digital converters use $2^n - 1$ comparators to convert an analog input to a digital value for processing. Flash ADCs are a series of comparators, each with a slightly different reference voltage. The priority encoder produces an output equal to the highest value input.



➤ In IC flash converters, the priority encoder usually includes a latch that holds the converter data constant for a period of time after the conversion.

➤ Fast – but more expensive

Example:

A certain 8-bit DAC has a full-scale output of 2mA and a full-scale error of $\pm 0.5\%$ F.S. What is the range of possible outputs for an input of 10000000?

Solution

The step size is $2\text{mA}/256 = 7.81 \mu\text{A}$. Since $10000000 = 128_{10}$, the ideal output should be:

$$128 \times 7.81 \mu\text{A} \approx 1000 \mu\text{A}.$$

The error can be as much as $\pm 0.5\% \times 2\text{mA} = \pm 10\mu\text{A}$

Thus, the actual output can deviate by this amount from the ideal $1000 \mu\text{A}$, so the actual output can be anywhere from 990 to $1010 \mu\text{A}$.

Problem 1:

Calculate the full-scale output voltage of a 5 bit binary ladder, if we consider the two logic levels as $1 = 20\text{V}$, $0 = 0\text{V}$ and $R_f = R$. **Ans: -19.375V**

Problem 2:

Calculate the analog output voltage at the given input combination (1010) of 4-bit binary ladder, if $0 = 0\text{V}$, $1 = 16\text{V}$ and $R_f = R$. **Ans: -10V**

$$V_{out} = \frac{R_f V_r}{R} (b_1 2^{-1} + b_2 2^{-2} + b_3 2^{-3} + b_4 2^{-4})$$

Problem 3

A 6-bit video D/A converter produces a maximum output of 1.25V. Determine the output voltage for the following digital input words:

a) 000001.

b) 100000.

c) 111111.

d) 011101

Solution

$$\text{Step size} = \frac{V_{fs}}{2^n} = \frac{V_{fs}}{2^6} = \frac{1.25V}{64} = 19.53mV$$

- a. output voltage = $1 \times 19.53\text{mV} = 19.53\text{mV}$
- b. output voltage = $32 \times 19.53\text{mV} = 624.96 \text{ mV}$
- c. output voltage = $63 \times 19.53\text{mV} = 1.23039 \text{ V}$
- d. output voltage = $29 \times 19.53\text{mV} = 566.32\text{mV}$

Problem 4

A 10-bit instrumentation D/A converter produces an output of 16mV with an input of 0000000100. Determine:

- a) The step size.
- b) The maximum output signal.

Solution

$$\text{Step size} = \frac{16\text{mV}}{4} = 4\text{ mV}$$

$$\text{Maximum output} = 1023 \times 4\text{ mV} = 4.092\text{V}$$

Problem 5

A certain 5-bit D/A converter has $V_{fs} = 10\text{V}$, with a guaranteed accuracy of $\pm 1/2$ LSB. Determine the expected (ideal) output, the upper and lower limits of the possible output and the percent error of the output at each of the following binary inputs:

- a) 00001.
- b) 00010.
- c) 10000.

Solution

$$1\text{LSB} = \frac{V_{fs}}{2^5} = \frac{10V}{32} \equiv 312.5\text{mV}$$

$$\pm \frac{1}{2} \text{ LSB} = 156.25\text{mV}$$

a. Expect output = $1 \times 312.5\text{mV} = 312.5\text{mV}$

Upper output limit = $312.5 \text{ mV} + 1/2 \text{ LSB}$

$$= 312.5\text{mV} + 156.25\text{mV} = 468.75\text{mV}$$

Lower output limit = $312.5 \text{ mV} - 1/2 \text{ LSB}$

$$= 312.5\text{mV} - 156.25\text{mV} = 156.25 \text{ mV}$$

Maximum output error = ± 50 percent

b. Expect output = $2 \times 312.5\text{mV} = 625 \text{ mV}$

Upper output limit = $625 \text{ mV} + 1/2 \text{ LSB}$

$$= 625 \text{ mV} + 156.25\text{mV} = 781.25\text{mV}$$

Lower output limit = $625 \text{ mV} - 1/2 \text{ LSB}$

$$= 625 \text{ mV} - 156.25\text{mV} = 468.75 \text{ mV}$$

Maximum output error = ± 25 percent

c. Expect output = $16 \times 312.5\text{mV} = 5 \text{ V}$

Upper output limit = $5\text{V} + 1/2 \text{ LSB}$

$$= 5\text{V} + 156.25\text{mV} = 5.156\text{V}$$

Lower output limit = $5\text{V} - 1/2 \text{ LSB}$

$$= 5\text{V} - 156.25\text{mV} = 4.844 \text{ mv}$$

Maximum output error = ± 3.12 percent

Problem 6

Consider a 10-bit dual slope analog-to-digital converter, with reference voltage of -1V, and a clock frequency of 1 KHz. The integrating capacitor charges to -8V in the time t_1 to t_2 , and discharges at a rate of 10V/sec in the time t_2 to t_3 . What is the input voltage if the binary reading is 1100100000.

Solution

$$T_{clock} = \frac{1}{f_{clock}} = \frac{1}{1 \times 10^3} = 1ms$$

$$T_1 = 2^{10} \times 1ms = 1024ms$$

$$T_2 = 800 \times 1ms = 800ms$$

$$\frac{T_2}{T_1} = \frac{V_{in}}{V_{ref}}$$

$$V_{in} = V_{ref} \frac{T_2}{T_1} = 1V \times \frac{800ms}{1024ms} \equiv 0.78V$$